HW01-AI Main Report

Software Testing

Field	Value
Full Name	Trần Hải Đức
Student ID	23127173
Assignment Code	HW01-AI
Course	Software Testing
Instructors	Dr. Lam Quang Vu Dr. Tran Duy Hoang MSc. Tran Thi Bich Hanh MSc. Truong Phuoc Loc MSc. Ho Tuan Thanh
Report Date	2026-05-31
AI Policy	Open – mandatory declaration and AI Audit Report attachment

Ho Chi Minh City, May 2026

0 Contents

1 General Information

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Full Name	Trần Hải Đức
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2 Submitted Artifact Summary

Artifact Group	Path
Requirement 1 report	`HW1/23127173/requirement/requirement1/requirement1.md`
Requirement 1 job data	`HW1/23127173/requirement/requirement1/jobs-data/`
Requirement 1 AI Mindmap Review	`HW1/23127173/requirement/requirement1/AI-Mindmap-Review/ai-mindmap-review.md`
Requirement 2 report	`HW1/23127173/requirement/requirement2/requirement2.md`
Requirement 3 report	`HW1/23127173/requirement/requirement3/requirement3.md`
Requirement 3 device evidence	`HW1/23127173/requirement/requirement3/devices/devices.jpg`
Requirement 3 video links	`HW1/23127173/requirement/requirement3/video-youtube-test/link-video.md`
Requirement 3 AI screenshot	`HW1/23127173/requirement/requirement3/screenshot-AI/screenshot-chat-ai.png`
Requirement 3 GitHub Issues	`HW1/23127173/requirement/requirement3/github-issues/`
Requirement 3 GitHub screenshots	`HW1/23127173/requirement/requirement3/github-issues/screenshot-defect/`
AI Audit Report	`HW1/23127173/doc/md/AI Audit/01_AI-Audit-Report.md`
AI Critique	`HW1/23127173/doc/md/AI Audit/02_AI-Critique.md`
Mandatory Disclosure	`HW1/23127173/doc/md/AI Audit/03_Mandatory-Disclosure.md`
AI Privacy Checklist	`HW1/23127173/doc/md/AI Audit/04_AI-Privacy-Checklist.md`
Prompt Log	`HW1/23127173/doc/md/appendixA-prompt-log.md`

Artifact Group	Path
Submission Checklist	`HW1/23127173/checklist.md`

3 Requirement 1 – QA/QC Job Market 2026+

Requirement 1 collected **10 QA/QC job postings** within 60 days prior to the collection date of **2026-05-28**. The dataset consists of 10 Markdown job-detail files, 10 screenshots, and one summary CSV file.

Criterion	Status
10 QA/QC job postings collected	Achieved
Each posting has a source link	Achieved
Each posting has a screenshot	Achieved
Each posting has job description and required skills	Achieved
Each posting has salary or explicit "not disclosed" note	Partially Achieved
$\geq$3 postings mention AI / LLM / Automation-AI	Achieved (Jobs 02, 03, 04, 06, 08)
Each posting has an AI Impact Analysis	Achieved

Key Observation: The 2026+ QA/QC market extends well beyond traditional manual and automation testing to encompass SDET, AI/LLM testing, data testing, Salesforce/AI solution QC, device/mobile QA, and design verification. AI tools assist in test generation, log analysis, documentation, and metric aggregation; however, QA engineers remain responsible for verifying results, protecting data, and owning release decisions.

4 Requirement 2 – 20 Software Defects (2022–2026)

Requirement 2 compiles **20 publicly documented software defects** from 2022 to 2026. Each defect entry includes a source link, description, severity, consequences, remediation, and a note on how an AI might produce biased or hallucinated explanations for that defect.

Criterion	Status
20 publicly known software defects	Achieved
All defects fall within the 2022–2026 period	Achieved
Source link provided for each defect	Achieved
Description / severity / consequences / remediation included	Achieved
$\geq$5 defects related to AI/LLM	Achieved (9 out of 20)
All 20 entries include AI bias/hallucination notes	Achieved

Key Observation: Defects from 2022–2026 highlight that QA/QC must address security vulnerabilities, cloud misconfiguration, supply-chain attacks, service outages, data privacy breaches, and AI behavioural failures. For AI/LLM systems, bugs may reside in prompts, guardrails, retrieval data, tool access permissions, and model reasoning paths.

5 Requirement 3 – Test Cases for a Physical Product

5.1 Device Under Test

The selected device is the **Casper Remote U25 Series air-conditioner remote control**. A photo of the remote together with the student ID card is provided in `devices/devices.jpg`.

Field	Value
Device	Casper Remote U25 Series Air-Conditioner Remote
Photo with ID	Achieved
Brand/Model	Achieved
Year / Serial	Not visible on provided device evidence
Number of test cases	18
Number of evidence videos	5 video links
AI-missed edge cases	TC-16, TC-17, TC-18
Confirmed defects	4 defects from TC-08, TC-13, TC-14, TC-16
GitHub Issues	4 real issues in `HappyDuckCoder/Software-Testing`; username screenshots are stored in `github-issues/screenshot-defect/`

5.2 Student-Discovered Edge Cases

The following three edge cases were identified by the student after reviewing AI-generated test cases:

TC	Description	Verdict
TC-16	The air conditioner can only be powered on via the Power button; however, Turbo, Mode, and Speed buttons can independently turn the unit on.	Fail
TC-17	Switching from Cool to Dry mode automatically reduces fan speed to the lowest setting.	Pass
TC-18	Baby Care mode locks configuration parameters such as temperature and airflow direction.	Pass

5.3 Confirmed Defects

Note: The requirement target is ≥5 confirmed defects; currently only 4 have been confirmed. No defects were fabricated. If the target must be met, additional testing must be conducted and defects added only upon real evidence.

Bug screenshots policy: HW01 does not use FIT Mantis. The four confirmed defects have been logged as GitHub Issues #1 to #4 in the personal repository `HappyDuckCoder/Software-Te`

Defect ID	Related TC	Summary
D-01	TC-08	iSAVE does not save or update the most recent configuration.
D-02	TC-13	Remote continues to operate even when the IR emitter is physically blocked.
D-03	TC-14	Air conditioner accepts commands at all tested angles of inclination.
D-04	TC-16	Turbo / Mode / Speed buttons can power on the unit while the air conditioner is off.

The issue links are recorded in `github-issues/github-issues-links.md`. The issue-list screenshot and individual issue screenshots are stored in `github-issues/screenshot-defect/` and show the GitHub username `HappyDuckCoder`.

6 AI CLO / Bloom-AI Outcomes

CLO	Requirement	Evidence	Status
G9.1 – Understand	Request AI to generate a QA/QC role mindmap, then identify errors.	`ai-mindmap-review.md`	Achieved
G9.3 – Analyse	Analyse AI output and find ≥3 edge cases the AI missed.	TC-16, TC-17, TC-18; `screenshot-chat-ai.png`	Achieved

In the mindmap review, the initial AI output omitted: Data Test Engineer, Design Verification, Device/Mobile QA, and the distinction between AI-assisted QA and AI/LLM system testing. The student confirmed these omissions as valid after cross-referencing all 10 jobs from Requirement 1 and produced a corrected mindmap.

7 AI Audit Summary

Metric	Result
Total AI-generated artifacts audited	17
VALID	0
INVALID	0
INCOMPLETE	17

All AI-generated artifacts are marked **INCOMPLETE** because AI was used solely to create drafts, standardise formatting, aggregate data, and offer suggestions. The student must verify every artifact against primary evidence: job screenshots, source links, videos, device photos, actual results, and defect evidence.

8 AI Critique

In HW01, AI was most useful in an editorial and structural role: standardising 10 Job Descriptions from Requirement 1 into Markdown/CSV, aggregating the 20 software defects for Requirement 2, proposing test cases for the air-conditioner remote in Requirement 3, scaffolding the G9.1 mindmap, drafting an English LaTeX version of the final report, converting the confirmed defect log into local GitHub Issue drafts, and updating the report after the student added real GitHub Issues screenshots. However, the audit shows that all 17 artifacts should be considered **INCOMPLETE** until primary evidence is verified by the student. AI can write coherently, but it cannot view screenshots, guarantee that links remain valid, execute tests on real hardware, or substitute for prohibited evidence such as device photos, videos, actual results, real GitHub Issues, or defect proof.

AI errors fell into three main categories:

1. **Over-inference:** When job descriptions were ambiguous, AI inferred AI skills for generic automation roles or interpreted undisclosed salaries without sufficient basis.
2. **Representation bias:** AI's training skews toward web/software QA patterns, so the initial mindmap omitted data testing, design verification, device/mobile QA, and AI-agent safety. The student confirmed these omissions as valid after comparing against the 10 collected jobs.
3. **Untested physical states:** For the air-conditioner remote, AI generated well-structured button-press test cases but overlooked physical preconditions such as the unit being powered off (TC-16), mode-transition side effects (TC-17), and configuration-lock behaviour under Baby Care (TC-18).

Lesson learned: AI should not be treated as the final source of truth. The correct workflow is: let AI draft, separate AI output from primary evidence, cross-reference against screenshots/links/real hardware, record audit verdicts, and only conclude Pass/Fail/defect upon real-world observation. AI accelerates the process; quality assurance and academic integrity remain the student's responsibility.

9 Mandatory Disclosure

> *Requirement 1 dataset, Requirement 2 software-defect dataset, Requirement 3 test-design draft, summary tables, CSV, mindmap review, checklist, GitHub Issue draft text, draft report sections, and the English LaTeX report draft were generated and formatted with assistance from Codex / ChatGPT and Claude. I provided or corrected source job details for Requirement 1, reviewed the generated Requirement 2 defect list, sources, severity, consequences, fixes, and AI bias/hallucination notes, and reviewed/modified the Requirement 3 remote air-conditioner test cases. I added student-found edge cases TC-16, TC-17, and TC-18 about off-state feature buttons, Cool-to-Dry fan adjustment, and Baby Care fixed configuration. I used Claude with the*

prompt "cho toi ban latex bang tieng anh dua tren noi dung tren" to draft an English LaTeX version from my existing report content. The LinkedIn screenshots, source links, physical device photo, videos, actual results, real GitHub Issues, GitHub Issues screenshot, and defect evidence are collected or verified manually by me. The detailed AI Audit Report is attached in `doc/md/AI Audit/01_AI-Audit-Report.md`*. I confirm I did not use AI to generate any prohibited artifact, including job posting screenshots with account name, physical device photos, videos, GitHub Issues screenshots, or fake evidence.*

10 Privacy and Responsible AI Use

- ✓ The assignment permits AI use under an Open policy, with mandatory declaration.
- ✓ All key prompts are logged in `appendixA-prompt-log.md`.
- ✓ Every AI-assisted artifact is listed in the AI Audit Report.
- ✓ AI was not used to generate job screenshots, device photos with student ID, test videos, actual results, or defect evidence.
- ✓ AI-assisted GitHub Issue drafts are only local draft text; real issues and the username screenshot must be created and verified manually by the student.
- ✓ All unverifiable fields are explicitly marked `Not disclosed` or `Not visible`.
- ✓ Final checks for YouTube video access, source links, screenshot account names, and GitHub Issues are pending before submission.

11 Self-Assessment

12 Declaration and Signature

I hereby declare that all prohibited evidence – including job posting screenshots with account name, device photos with student ID, test videos, actual results, and defect evidence – was not generated by AI. All AI-assisted sections have been declared in the AI Audit Report and Appendix A Prompt Log.

Item	Rubric Area	Max	Self	Note
1	Job Market 2026+	40	38	10 jobs, screenshots, sources, CSV, and AI impact are included; salary/not-disclosed fields and screenshots need a final manual check.
2	Software Defects 2022–2026	20	20	20 public defects with source, severity, consequence, remediation, and AI bias/hallucination note.
3	Physical-product test design	25	22	Device photo, 18 test cases, 5 videos, GitHub Issues, and username screenshots are included; main risk is only 4 confirmed defects instead of the target ≥5.
AI-1	AI Audit Report	8	8	AI Audit Report is included.
AI-2	AI Critique + Disclosure	4	4	Critique and mandatory disclosure are included.
AI-3	AI Checklist + anti-cheat artifacts	3	3	Checklist, privacy checklist, prompt log, and real evidence artifacts are included.
Total		**100**	**95**	**Proposed self-score: 95 / 100.**

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Full Name	Trần Hải Đức
Student ID	23127173
Date Signed	2026-05-31
Signature	*Trần Hải Đức*